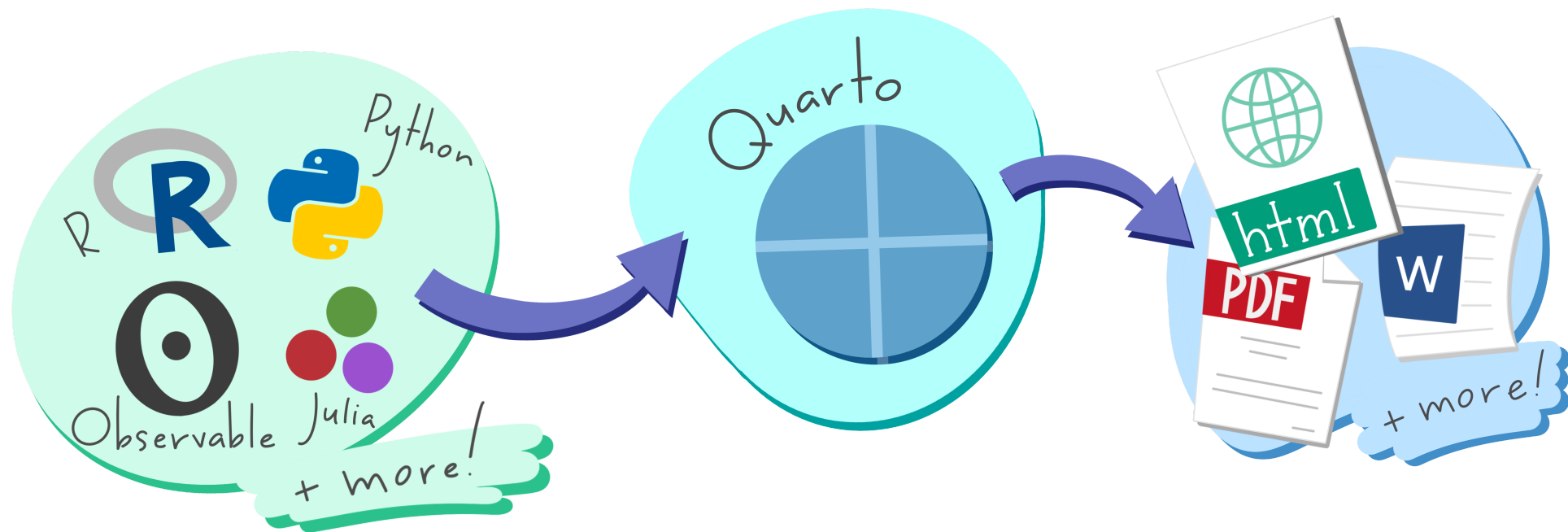


Dynamic documents in R

reproducible research with Quarto

2026-04-23



Artwork from “Hello, Quarto” keynote by Julia Lowndes and Mine Çetinkaya-Rundel, presented at RStudio Conference 2022. Illustrated by Allison Horst.

```
---
title: "ggplot2 demo"
author: "Norah Jones"
date: "5/22/2021"
format:
  html:
    fig-width: 8
    fig-height: 4
    code-fold: true
---
```

Source

```
## Air Quality
```

```
@fig-airquality further explores the impact of
temperature on ozone level.
```

```
```{r}
#| label: fig-airquality
#| fig-cap: Temperature and ozone level.
#| warning: false
```

```
library(ggplot2)
```

```
ggplot(airquality, aes(Temp, Ozone)) +
 geom_point() +
 geom_smooth(method = "loess")
)
```
```



ggplot2 demo

Norah Jones
May 22nd, 2021

Output

Air Quality

[Figure 1](#) further explores the impact of temperature on ozone level.

► Code

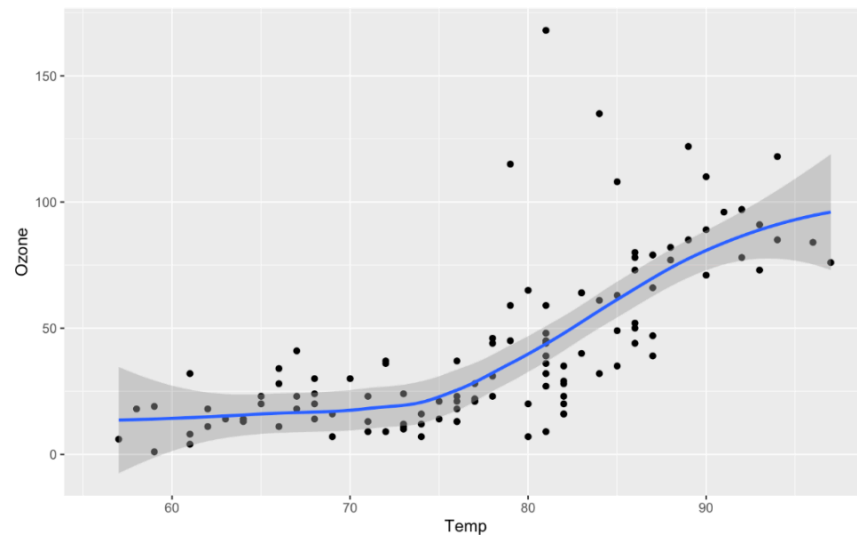
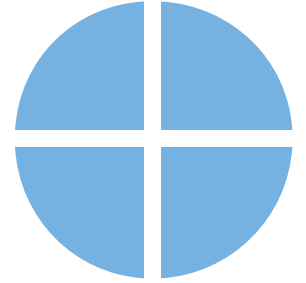


Figure 1: Temperature and ozone level.

Quarto

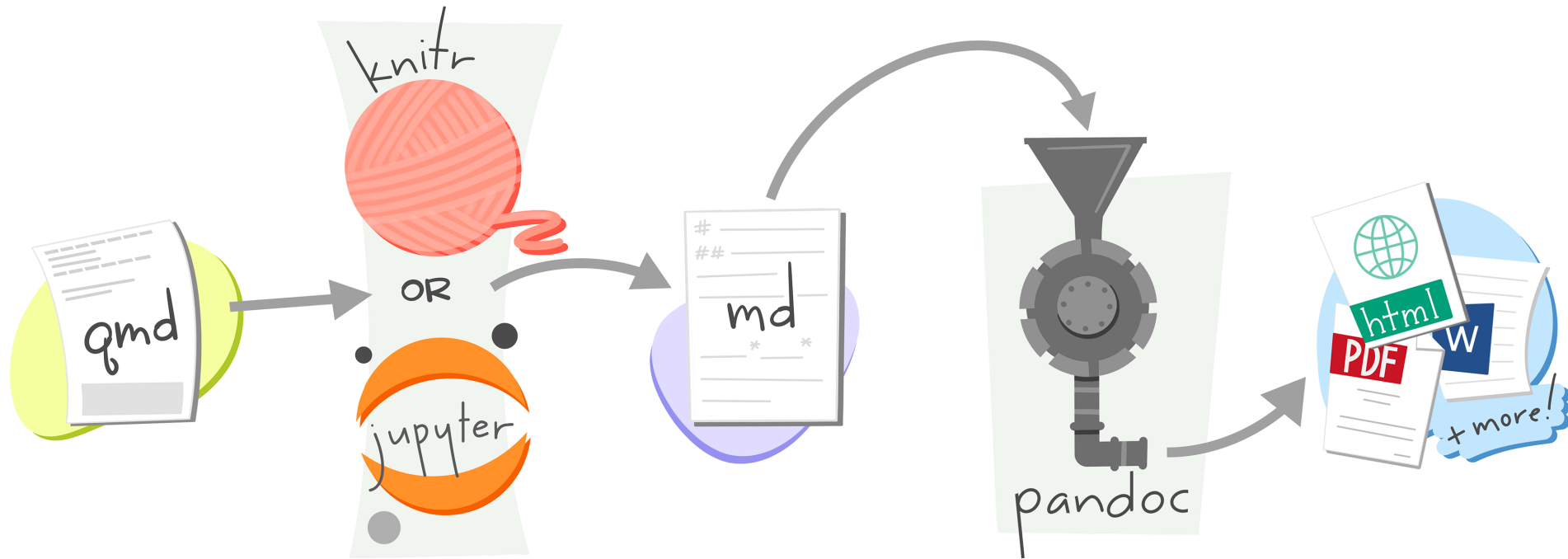


Dynamic: code and text in same document

Reproducible: re-run your analysis, re-render your document

Flexible: output to different formats easily

Rendering



Your Turn 1

Create a new Quarto file. Go to File > New File > Quarto Document. Press OK. Save the file and press the “Render” button.



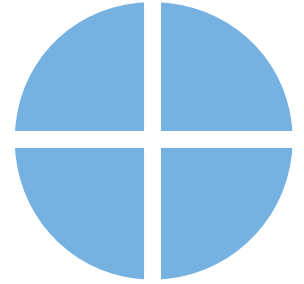
```
1 ---
2 title: "Untitled"
3 format: html
4 ---
5
6 ## Quarto
7
8 Quarto enables you to weave together content and executable code into a
9 finished document.
10 To learn more about Quarto see <https://quarto.org>.
11 ## Running Code
12
13 When you click the Render button a document will be generated that includes
14 both content and the output of embedded code.
15 You can embed code like this:
16 ```{r}
17 1 + 1
18 ```
19
20 You can add options to executable code like this
21
22 ```{r}
23 #| echo: false
24 2 * 2
25 ```
26
27 The `echo: false` option disables the printing of code (only output is
28 displayed).
```

document metadata

plain text

code chunks

Quarto
Prose
Code
Metadata

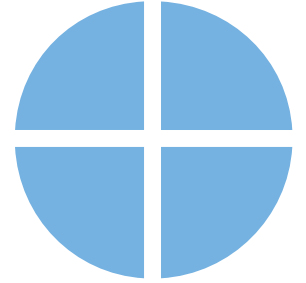


Quarto

Prose = *Markdown*

Code

Metadata



the Visual Editor

relational-data.qmd

Render on Save Render Run

Source Visual B I <> Normal Format Insert Table Outline

#filtering-joins

Filtering Joins

Filtering joins match observations in the same way as [mutating joins](#), but affect the observations, not the variables. There are two types:

| | | |
|------------------------------|----------------------|---|
| <code>semi_join(x, y)</code> | $x \ltimes y$ | Keeps all observations in <code>x</code> that have a match in <code>y</code> |
| <code>anti_join(x, y)</code> | $x \triangleright y$ | Drops all observations in <code>x</code> that have a match in <code>y</code> |

Graphically, a semi-join looks like this:

```
{r:join-semi}-  
#l-echo: false-  
#l-out-width: "6"-  
-  
knitr::include_graphics("diagrams/join-semi.png")
```

| key | val_x |
|-----|-------|
| 1 | x1 |
| 2 | x2 |

Only the existence of a match is important; it doesn't matter which observation is matched. This means that filtering joins never duplicate rows like mutating joins do:

Chunk 1: join-semi

Quarto

Basic Markdown Syntax

`*italic*` `**bold**`

`_italic_` `__bold__`

Basic Markdown Syntax

Header 1

Header 2

Header 3

Basic Markdown Syntax

`<http://example.com>`

`[linked phrase](http://example.com)`

Learn more about Markdown Syntax with the ten-twenty minute tutorial on markdown at <https://commonmark.org/help/tutorial>.

Your Turn 2 (open **exercises.qmd**)

Read this short introduction to the Visual Editor:

<https://quarto.org/docs/visual-editor/>

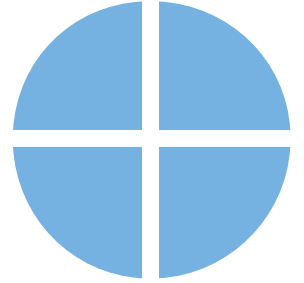
Use the Visual Editor to stylize the text from the Gapminder website below. Experiment with bolding, italicizing, making lists, etc.

Quarto

Prose

Code = *R code chunks*

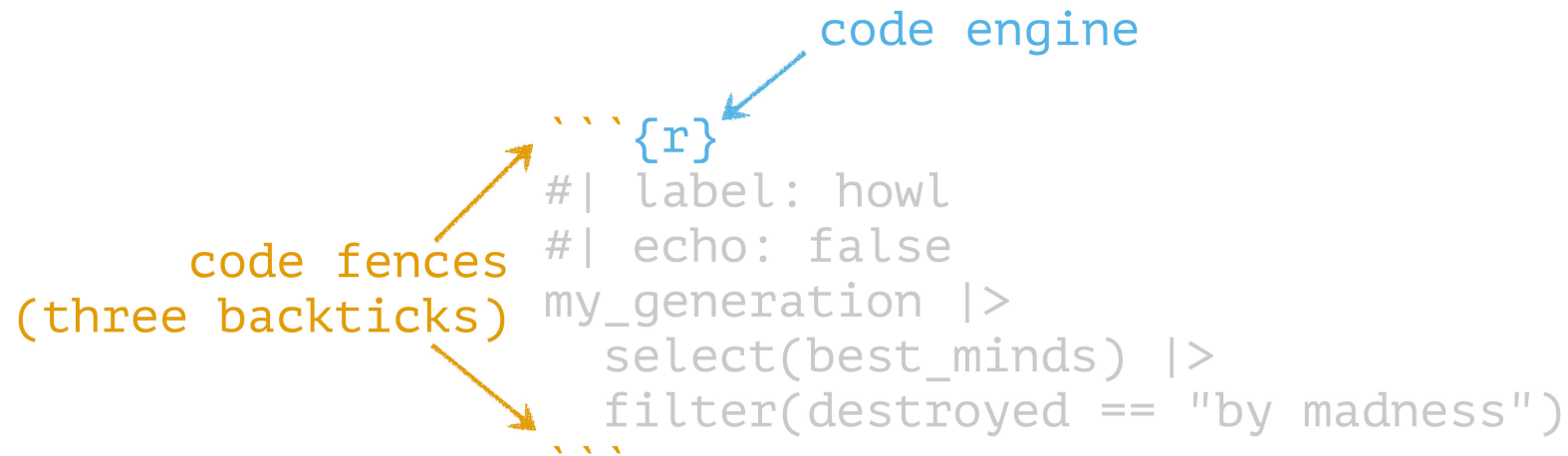
Metadata



Code chunks

```
` `` {r}  
#| label: howl  
#| echo: false  
my_generation |>  
  select(best_minds) |>  
  filter(destroyed == "by madness")  
` ``
```

Code chunks



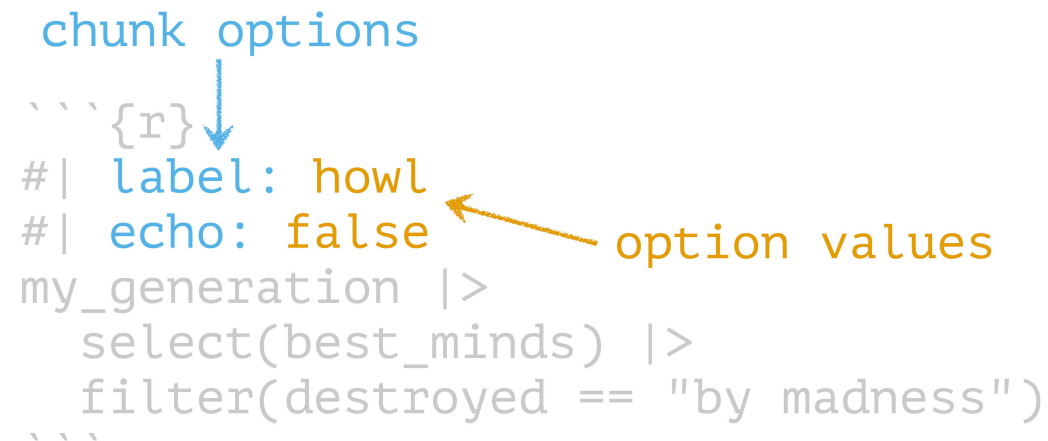
The diagram illustrates the components of a code chunk. It features a code block enclosed in three backticks. An orange arrow points from the text 'code fences (three backticks)' to the opening backticks. Another orange arrow points from the same text to the closing backticks. A blue arrow points from the text 'code engine' to the opening curly brace of the R code snippet.

```
code fences  
(three backticks) {r}  
#| label: howl  
#| echo: false  
my_generation |>  
  select(best_minds) |>  
  filter(destroyed == "by madness")
```

Code chunks

```
chunk options
````{r}
#| label: howl
#| echo: false
my_generation |>
 select(best_minds) |>
 filter(destroyed == "by madness")
````
```

option values



Code chunks

code chunk comments

↓
``{r}
#| label: howl
#| echo: false
my_generation |>
 select(best_minds) |>
 filter(destroyed == "by madness")
``

Chunk options

| Option | Effect |
|----------------------------------|--|
| <code>include: false</code> | run the code but don't print it or results |
| <code>eval: false</code> | don't evaluate the code |
| <code>echo: false</code> | run the code and output but don't print code |
| <code>message: false</code> | don't print messages (e.g. from a function) |
| <code>warning: false</code> | don't print warnings |
| <code>fig.cap: "Figure 1"</code> | caption output plot with "Figure 1" |

See the [Quarto documentation](#)

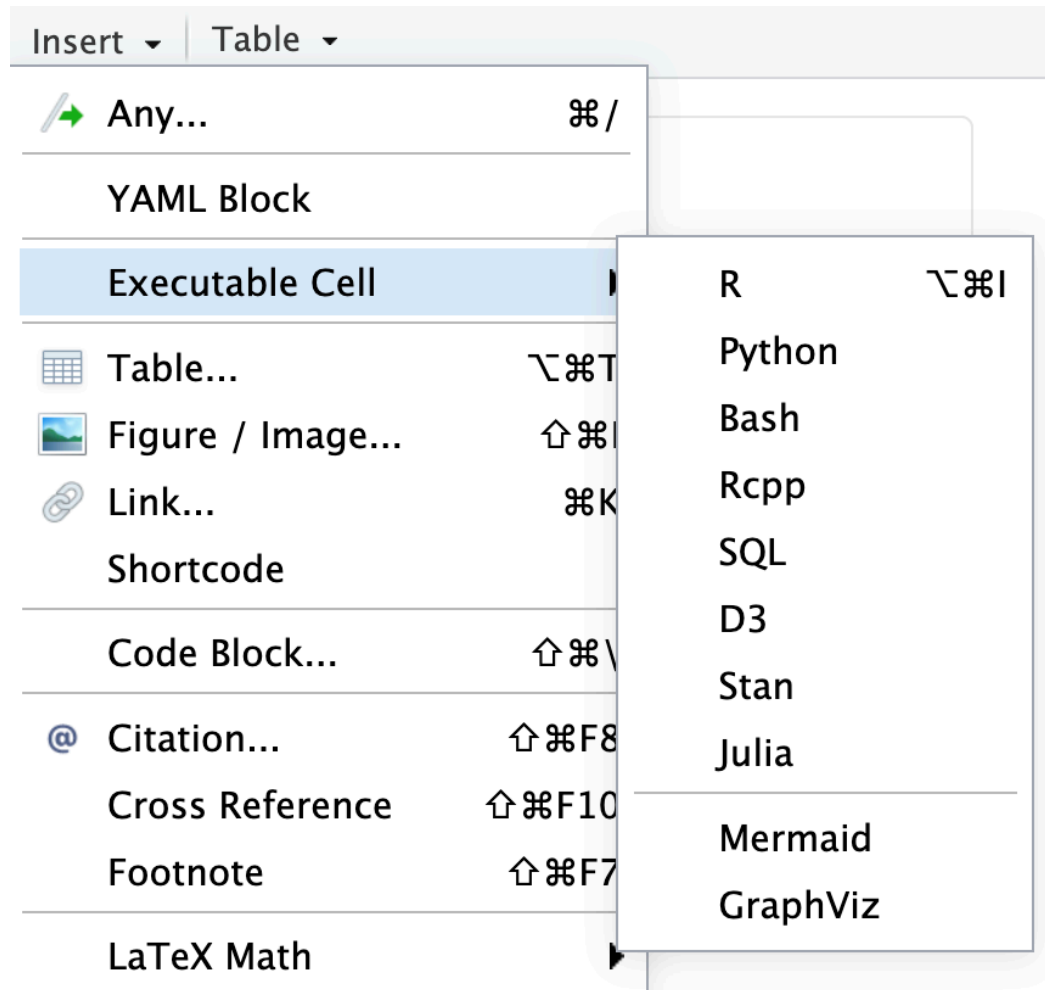
Other languages

First-class support for Python, Julia, and
Observable JS

Supports Jupyter notebooks

57 knitr engines

Insert code chunks with **cmd/ctrl + alt/option + I** or Visual Editor



Press **tab** to autocomplete chunk options

```
```{r}
#| echo: false
#|
#|
#|
knitr::i
##
##
Op
```

- label:
- classes:
- cache:
- cache-vars:
- cache-globals:
- cache-lazy:
- cache-rebuild:
- cache-comments:

**label**  
Unique label for code cell

You can type it in manually, use the keyboard short-cut (Cmd/  
"Insert" button above. Put the following code in it:

```
```{r}
#| echo:
#| out.h
#| out.w
knitr::i
```
```

- true
- false
- fenced

## *Your Turn 3*

**Create a code chunk. You can type it in manually, use the keyboard short-cut (Cmd/Ctrl + Option/Alt + I), or use the “Insert” button above. Put the following code in it:**

```
1 gapminder |>
2 slice(1:5) |>
3 gt()
```

**Render the document**

## *Your Turn 4*

**Add `echo: false` to the code chunk you created and re-render. What's the difference in the output?**

# Inline Code

```
who wept at the romance of the
streets with their pushcarts full of
onions and bad music,
who sat in boxes breathing in the
darkness under the bridge, and rose
up to build `r select(instruments,
harpsichord)` in their lofts,
who coughed on the `r length(floors)`
floor of Harlem crowned with flame
under the tubercular sky surrounded
by orange crates of theology,
who scribbled all night rocking and
rolling over lofty incantations which
in the yellow morning were stanzas of
gibberish,
```

# Inline Code

who wept at the romance of the  
streets with their pushcarts full of  
onions and bad music,  
who sat in boxes breathing in the  
darkness under the bridge, and rose  
up to build ``r select(instruments,`  
+ `r harpsichord)`` in their lofts,  
who coughed on the ``r length(floors)``  
floor of Harlem crowned with flame  
under the tubercular sky surrounded  
by orange crates of theology,  
who scribbled all night rocking and  
rolling over lofty incantations which  
in the yellow morning were stanzas of  
gibberish,

single  
backtick →  
+ r

any R code



## *Your Turn 5*

Remove `eval: false` so that Quarto evaluates the code.

Use `summarize()` and `n_distinct()` to get the the number of unique years in `gapminder` and save the results as `n_years`.

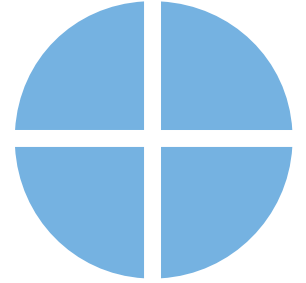
Use inline code to describe the data set in the text below the code chunk and re-render.

# Quarto

Prose

Code

**Metadata** = *YAML*



# YAML Metadata

```
1 ---
2 title: "Howl"
3 author: "Allen Ginsberg"
4 date: "March 18, 1956"
5 format:
6 html: default
7 pdf:
8 toc: true
9 number-sections: true
10 ---
```

# Output formats

| Format                | Outputs                      |
|-----------------------|------------------------------|
| <code>html</code>     | HTML                         |
| <code>pdf</code>      | PDF                          |
| <code>word</code>     | Word <code>.docx</code>      |
| <code>odt</code>      | OpenOffice <code>.odt</code> |
| <code>gfm</code>      | GitHub-flavored Markdown     |
| <code>revealjs</code> | Reveal Slides (HTML)         |
| <code>beamer</code>   | Beamer Slides (PDF)          |
| <code>pptx</code>     | Powerpoint Slides            |

## Your Turn 6

Set figure chunk options to the code chunk below, such as `dpi`, `fig.width`, and `fig.height`. Run `knitr::opts_chunk$get()` in the console to see the defaults.

Add your name to the YAML header using `author: Your Name`.

Change `format: html` to use the `toc: true` and `code-fold: true` options and re-render

# Parameters

```
1 ---
2 params:
3 param1: x
4 param2: y
5 data: df
6 ---
```

## *Calling parameters in R*

```
1 params$param1
2 params$param2
3 params$data
```

## *From the Console*

```
1 quarto::quarto_render(
2 "document.qmd",
3 execute_params = list(param1 = 0.2, param2 = 0.3)
4)
```

## Your Turn 7

Change the **params** option in the YAML header to use a different continent. Re-render.

```
1 gapminder |>
2 filter(continent == params$continent) |>
3 ggplot(aes(x = year, y = lifeExp, group = country, color = country)) +
4 geom_line(lwd = 1, show.legend = FALSE) +
5 scale_color_manual(values = country_colors) +
6 theme_minimal(14) +
7 theme(strip.text = element_text(size = rel(1.1))) +
8 ggtitle(paste("Continent:", params$continent))
```

# Bibliographies and citations

*Bibliography files: .bib, Zotero, others*

*Citation styles: .csl*

*[@citation-label]*

*Visual Editor's citation wizard can help!*



# Including bibliography files in YAML

```
1 ---
2 bibliography: file.bib
3 csl: file.csl
4 ---
```

*the Visual Editor can also manage this for you.*

# Your turn 8

**Cite Causal Inference in text below. Using the citation wizard, find the right citation under My sources > Bibliography.**

**Add the American Journal of Epidemiology CSL to the YAML using csl: [aje.csl](#)**

**Re-render**

# Make cool stuff in Quarto!

- Books
- Blogs
- These slides!

See the Gallery for inspiration

# Resources

**Quarto Documentation:** A comprehensive but friendly introduction to Quarto. Written in Quarto!

**R for Data Science:** A comprehensive but friendly introduction to the tidyverse. Free online. Written in Quarto!

**Posit Recipes:** Common code patterns in R (with some comparisons to SAS)